

ON A PROBLEM OF ERDŐS AND SÁRKÖZY ON ODD CYCLES IN COPRIME GRAPHS

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ABSTRACT. We prove that if n is large, $A \subseteq \{1, \dots, n\}$ has more than $\lfloor n/2 \rfloor + \lfloor n/3 \rfloor - \lfloor n/6 \rfloor$ elements, and A omits at most $n/70$ of the integers $w \equiv 2, 4 \pmod{6}$, then the coprime graph of A contains every odd cycle of length at most $n/3 + 1$. This answers the question of Erdős and Sárközy (Erdős Problem #883, Question 1) for all sets of this kind, which include the extremal examples. The case in which more than $n/70$ of these integers are missing remains open.

1. INTRODUCTION

For $A \subseteq [n] := \{1, \dots, n\}$ let $G(A)$ denote the *coprime graph* of A : its vertex set is A and $x \neq y$ are adjacent if $\gcd(x, y) = 1$. Put

$$T(n) = \left\lfloor \frac{n}{2} \right\rfloor + \left\lfloor \frac{n}{3} \right\rfloor - \left\lfloor \frac{n}{6} \right\rfloor, \quad o(n) = \left\lfloor \frac{n}{3} \right\rfloor - \left\lfloor \frac{n}{6} \right\rfloor,$$

so that $T(n)$ is the number of integers in $[n]$ divisible by 2 or 3, and $o(n)$ is the number of odd multiples of 3 in $[n]$. The set of multiples of 2 or 3 shows that $|A| \leq T(n)$ does not force a triangle. Erdős and Sárközy [2] proved that $|A| > T(n)$ forces all odd cycles C_{2l+1} with $l \leq cn$, for a small absolute constant $c > 0$ and $n \geq n_0$, and asked whether the range can be extended to all lengths at most $n/3 + 1$. This is Question 1 of Erdős Problem #883 [1]. The bound would be best possible: if A consists of all even numbers together with $o(n) + 1$ odd numbers, then, since even numbers are pairwise non-adjacent, a cycle with L vertices contains at least $\lceil L/2 \rceil$ odd vertices, so every odd cycle has length at most $2o(n) + 1$.

The present note settles the question in the regime that contains these extremal configurations.

Theorem 1.1. *There is n_0 such that the following holds for every $n \geq n_0$. Let $A \subseteq [n]$ satisfy $|A| \geq T(n) + 1$, and suppose that at most $n/70$ integers $w \in [n]$ with $w \equiv 2, 4 \pmod{6}$ are missing from A . Then $G(A)$ contains a cycle of length $2l + 1$ for every integer l with $1 \leq l \leq o(n)$. In particular it contains every odd cycle of length at most $n/3 + 1$.*

Since $o(n) \geq n/6 - 1$, every odd length up to $n/3 - 1$ is covered, and every odd length $L \leq n/3 + 1$ satisfies $(L - 1)/2 \leq \lfloor n/6 \rfloor \leq o(n)$; so the last sentence follows from the first. The constant n_0 is effective; we do not attempt to compute it (all requirements on n are of the form “ n is large” in explicit elementary inequalities, see Section 5).

Remark 1.2. Theorem 1.1 does not settle Question 1 in full: a set A with $|A| > T(n)$ may omit many integers $w \equiv 2, 4 \pmod{6}$, and then the supply of even connectors shrinks while the number of odd elements grows. The remaining case is discussed in Section 6.

Strategy. Write O for the set of odd elements of A . Cycles of length at most 9 are supplied by the theorem of Erdős and Sárközy (Section 3). For long cycles we choose $l + 1$ odd vertices and look for a cycle that alternates between odd vertices and even numbers $w \equiv 2, 4 \pmod{6}$, except for exactly one edge joining two coprime odd vertices; such a cycle has length $2l + 1$. The alternating part is produced by Jackson’s theorem on cycles in bipartite graphs (Theorem 2.1), which needs every

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odd vertex to be coprime to more than half of the available even numbers. An odd z is coprime to a proportion $\rho'(z) = \prod_{p|z, p \geq 5} (1 - 1/p)$ of the numbers $w \equiv 2, 4 \pmod{6}$, so the odd numbers with $\rho'(z) < 0.56$ (“hard” numbers) must be treated separately; a moment estimate (Lemma 2.3) shows there are at most $6 \cdot 10^{-4}n$ of them, and each is embedded into a short path between two “very easy” odd numbers, which is then treated as a single vertex. Missing even numbers can destroy the neighbourhoods of hard numbers; we show that fewer such numbers are damaged than there are surplus odd elements, so they can simply be left out.

2. NOTATION AND TOOLS

Throughout, p denotes a prime. For a positive integer m let

$$\rho'(m) = \prod_{p|m, p \geq 5} \left(1 - \frac{1}{p}\right), \quad \omega(m) = \#\{p : p \mid m\}.$$

Clearly $\rho'(\text{lcm}(m_1, m_2)) \geq \rho'(m_1)\rho'(m_2)$, since every prime factor of the least common multiple occurs in m_1 or in m_2 and all factors are at most 1. Also $\rho'(m) \geq \varphi(m)/m$. Let

$$E^* = E^*(n) = \{w \in [n] : w \equiv 2 \text{ or } 4 \pmod{6}\}$$

be the set of even numbers in $[n]$ not divisible by 3; $|E^*| = \lfloor n/2 \rfloor - \lfloor n/6 \rfloor \in [n/3 - 1, n/3 + 1]$. For $m \geq 1$ put $N^*(m) = \{w \in E^* : \gcd(w, m) = 1\}$. Note that for odd z and $w \in E^*$, w and z are adjacent in $G([n])$ if and only if $w \in N^*(z)$; and $w \in N^*(z_1) \cap N^*(z_2)$ if and only if $w \in N^*(\text{lcm}(z_1, z_2))$.

2.1. Jackson’s theorem. For a bipartite graph with parts X, Y we use the following result of Jackson [3].

Theorem 2.1 (Jackson [3]). *Let H be a bipartite graph with parts X and Y , where $|X| \geq 2$, and let $k \geq 2$ be an integer such that every vertex of X has at least k neighbours in Y . If $|X| \leq k$ and $|Y| < 2k - 2$, then H contains a cycle of length $2|X|$; that is, a cycle through all vertices of X .*

(Jackson states the condition on Y as $|Y| \leq 2k - 2$ in some sources and $|Y| < 2k - 2$ in others; we only use it with $|Y| \leq 2k - 3$, which satisfies both.)

2.2. Elementary arithmetic estimates. We use two classical explicit bounds: for every integer $m \geq 3$,

$$(1) \quad \omega(m) \leq 1.3841 \frac{\log m}{\log \log m} \quad (\text{Robin [4]}),$$

$$(2) \quad \frac{\varphi(m)}{m} \geq \frac{1}{e^\gamma \log \log m + \frac{2.51}{\log \log m}} \quad (\text{Rosser–Schoenfeld [5]}).$$

Since $\log m / \log \log m$ increases for $m \geq e^e$, (1) gives, for n large and every $m \leq n^2$,

$$(3) \quad 2^{\omega(m)} \leq \exp \left(1.3841 \log 2 \cdot \frac{2 \log n}{\log(2 \log n)} \right) \leq \sqrt{n},$$

because the exponent is at most $\frac{1}{2} \log n$ as soon as $\log(2 \log n) \geq 3.84$. Similarly (2) gives, for every $m \leq n^2$ and n large,

$$(4) \quad \rho'(m) \geq r(n) := \frac{1}{2 \log \log n + 3}.$$

Lemma 2.2 (Counting). *For every $m \geq 1$,*

$$|N^*(m)| > \frac{n}{3} \rho'(m) - 2^{\omega(m)}.$$

In particular, for n large and every $m \leq n^2$, $|N^(m)| > \frac{n}{3} \rho'(m) - \sqrt{n}$.*

Proof. Let m' be the product of the primes $p \geq 5$ dividing m . An element $w \in E^*$ is coprime to m if and only if it is coprime to m' (it is even and not divisible by 3, and 2, 3 are the only other possible common primes). For a squarefree d with $\gcd(d, 6) = 1$, the number of $w \in E^*$ with $d \mid w$ equals

$$\#\{w \leq n : 2d \mid w\} - \#\{w \leq n : 6d \mid w\} = \left\lfloor \frac{n}{2d} \right\rfloor - \left\lfloor \frac{n}{6d} \right\rfloor = \frac{n}{3d} + \theta_d, \quad |\theta_d| < 1.$$

By inclusion-exclusion,

$$|N^*(m)| = \sum_{d|m'} \mu(d) \left(\frac{n}{3d} + \theta_d \right) = \frac{n}{3} \prod_{p|m'} \left(1 - \frac{1}{p} \right) + \sum_{d|m'} \mu(d) \theta_d,$$

and the last sum has absolute value less than $2^{\omega(m')} \leq 2^{\omega(m)}$. The second statement follows from (3). $\square$

2.3. How many odd numbers are hard.

Lemma 2.3 (Moment bound). *For $0 < x \leq 1$ and every integer $s \geq 1$, let $\mathcal{F}_s(x) = \{z \leq n : \prod_{p|z, p \geq s} (1 - 1/p) \leq x\}$. Then for every integer $m \geq 1$,*

$$|\mathcal{F}_s(x)| \leq n x^m \Pi_{s,m}, \quad \Pi_{s,m} := \prod_{p \geq \max(s, 5)} \left(1 + \frac{(1 - 1/p)^{-m} - 1}{p} \right).$$

Numerically,

$$0.56^{40} \Pi_{5,40} < 6 \cdot 10^{-4}, \quad 0.75^{60} \Pi_{11,60} < 4 \cdot 10^{-4}.$$

Proof. Let g be the multiplicative function supported on squarefree integers all of whose prime factors are $\geq \max(s, 5)$, with $g(p) = (1 - 1/p)^{-m} - 1 \geq 0$ for such primes. Then $h = 1 * g$ satisfies $h(z) = \prod_{p|z, p \geq \max(s, 5)} (1 - 1/p)^{-m}$, so $h(z) \geq x^{-m}$ for $z \in \mathcal{F}_s(x)$ when $s \geq 5$ (we only use $s = 5$, where the product is $\rho'(z)$, and $s = 11$). Hence

$$x^{-m} |\mathcal{F}_s(x)| \leq \sum_{z \leq n} h(z) = \sum_{d \leq n} g(d) \left\lfloor \frac{n}{d} \right\rfloor \leq n \sum_{d \geq 1} \frac{g(d)}{d} = n \Pi_{s,m}.$$

For the numerical values we computed the finite product over primes $p < P_0 = 10^7$ in double precision (the relative rounding error is below 10^{-9}). For $p \geq P_0$ we have $(1 - 1/p)^{-m} \leq e^{m/(p-1)}$ and $e^t - 1 \leq 2t$ for $0 \leq t \leq 1$, so $g(p) \leq 2m/(p-1)$ and

$$\sum_{p \geq P_0} \log \left(1 + \frac{g(p)}{p} \right) \leq \sum_{j \geq P_0} \frac{2m}{j(j-1)} = \frac{2m}{P_0 - 1}.$$

This gives $0.56^{40} \Pi_{5,40} \leq 5.82 \cdot 10^{-4}$ and $0.75^{60} \Pi_{11,60} \leq 3.09 \cdot 10^{-4}$. $\square$

Definition 2.4. An odd number $z \leq n$ is *hard* if $\rho'(z) < 0.56$, *ordinary* if $\rho'(z) \geq 0.56$, and *easy* if $\rho'(z) \geq 0.75$.

Corollary 2.5. *For n large:*

- (a) *for every $x \leq 0.56$ the number of $z \leq n$ with $\rho'(z) \leq x$ is at most $1.04 \cdot 10^{-3} x n$; in particular there are at most $5.82 \cdot 10^{-4} n$ hard numbers;*
- (b) *the number of odd $z \leq n$ that are not easy is at most $0.1578 n$.*

Proof. (a) By Lemma 2.3 with $s = 5$, $m = 40$, the number is at most $n(x/0.56)^{40} \cdot 0.56^{40} \Pi_{5,40} \leq 5.82 \cdot 10^{-4} n (x/0.56)$, and $5.82 \cdot 10^{-4}/0.56 < 1.04 \cdot 10^{-3}$. (b) If z is odd and $\rho'(z) < 0.75$, then either $5 \mid z$ or $7 \mid z$, or z is coprime to 35, in which case $\prod_{p|z, p \geq 11} (1 - 1/p) = \rho'(z) < 0.75$. The odd multiples of 5 or 7 in $[n]$ number at most $\frac{n}{10} + \frac{n}{14} - \frac{n}{70} + 3 = \frac{11n}{70} + 3$, and by Lemma 2.3 with $s = 11$, $m = 60$ the remaining ones number at most $3.09 \cdot 10^{-4} n$. Since $11/70 < 0.15715$, the total is below $0.1578n$ for large n . $\square$

2.4. A coprime pair.

Lemma 2.6. *Let $n \geq 13$ and let O be a set of odd numbers in $[n]$ with $|O| \geq o(n) + 1$. Then O contains two coprime elements.*

Proof. Partition the odd numbers of $[n]$ into the groups $Q_k = \{6k + 1, 6k + 3, 6k + 5\} \cap [n]$, $0 \leq k \leq K := \lfloor (n-1)/6 \rfloor$. Any two elements of one group differ by 2 or 4, hence are coprime (a common divisor divides the difference and is odd). The number of groups is $K + 1$, while $o(n) = \#\{k \geq 0 : 6k + 3 \leq n\} = \lfloor (n-3)/6 \rfloor + 1 \geq K$. If $|O| > K + 1$, two elements lie in a common group and we are done. Otherwise $|O| = K + 1 = o(n) + 1$, which forces $\lfloor (n-3)/6 \rfloor = \lfloor (n-1)/6 \rfloor - 1$, i.e. $6K + 1 \leq n < 6K + 3$; then $Q_K = \{6K + 1\}$ and O meets every group in exactly one element, so $6K + 1 \in O$. Suppose O has no coprime pair. Its element in $Q_0 = \{1, 3, 5\}$ is not 1. Since $n \geq 13$, $Q_1 = \{7, 9, 11\}$, and its element must share a prime with the element of Q_0 ; the only possibility is the pair 3, 9. But $\gcd(3, 6K + 1) = 1$ and $6K + 1 \neq 3$, a contradiction. $\square$

3. PROOF OF THEOREM 1.1: SET-UP AND SHORT CYCLES

Fix $n \geq n_0$ and A as in Theorem 1.1. Let O be the set of odd elements of A , let d be the number of even numbers of $[n]$ missing from A , and let $D = E^* \setminus A$, $d_E = |D| \leq n/70$. Then $d \geq d_E$ and

$$(5) \quad |O| = |A| - (\lfloor n/2 \rfloor - d) \geq T(n) + 1 - \lfloor n/2 \rfloor + d = o(n) + 1 + d \geq o(n) + 1 + d_E.$$

Short cycles. Erdős and Sárközy [2, Theorem 1] proved that there are absolute constants $c > 0$ and n_1 such that for $n \geq n_1$ every $A \subseteq [n]$ with $|A| > T(n)$ (their $f(n, 2)$ is our $T(n)$) satisfies $C_{2l+1} \subseteq G(A)$ for all positive integers $l \leq cn$. Taking $n_0 \geq \max(n_1, 4/c)$ we obtain the cycles of length $2l + 1$ for $1 \leq l \leq 4$. It remains to treat $5 \leq l \leq o(n)$.

Damaged numbers. For odd $z \leq n$ put $a(z) = |N^*(z) \cap D|$, the number of its neighbours in E^* that are missing from A . Call a hard z *damaged* if $a(z) > \frac{n}{8}\rho'(z)$.

Lemma 3.1. *At most $0.0084 d_E$ odd numbers in $[n]$ are damaged. Consequently $O' = O \setminus \{\text{damaged}\}$ satisfies $|O'| \geq o(n) + 1 \geq n/6$.*

Proof. If z is damaged then $\frac{n}{8}\rho'(z) < a(z) \leq d_E$, so $\rho'(z) < 8d_E/n$; also $\rho'(z) < 0.56$. By Corollary 2.5(a) applied with $x = \min(8d_E/n, 0.56)$, the number of such z is at most $1.04 \cdot 10^{-3} \cdot (8d_E/n) \cdot n < 0.0084 d_E$. The second statement follows from (5). $\square$

By Lemma 2.6 applied to O' there are coprime $a, b \in O'$; we fix them. For every $z \in O'$ and every easy y ,

$$(6) \quad |N^*(\text{lcm}(y, z)) \cap A| \geq \frac{n}{8}\rho'(z) - \sqrt{n}.$$

Indeed $|N^*(\text{lcm}(y, z))| > \frac{n}{4}\rho'(z) - \sqrt{n}$ by Lemma 2.2, and at most $a(z)$ of these numbers are missing from A ; if z is hard then $a(z) \leq \frac{n}{8}\rho'(z)$, and if z is not hard then $a(z) \leq d_E \leq n/70 < \frac{n}{8} \cdot 0.56$.

4. PROOF OF THEOREM 1.1: LONG CYCLES

Now fix l with $5 \leq l \leq o(n)$. All estimates below hold once n is larger than an absolute constant; they are collected in Section 5.

4.1. Choice of the odd vertices. List the elements of $O' \setminus \{a, b\}$ so that the easy ones come first, then the ordinary ones that are not easy, then the hard ones (inside each class in any order). Let X_0 consist of a, b and the first $l - 1$ elements of this list; this is possible because $|O'| \geq o(n) + 1 \geq l + 1$ (Lemma 3.1), and $|X_0| = l + 1$. Let $z_1, \dots, z_h$ be the hard elements of $X_0 \setminus \{a, b\}$, ordered so that $\rho'(z_1) \leq \dots \leq \rho'(z_h)$. By Corollary 2.5(a), $h \leq 5.82 \cdot 10^{-4}n$. No element of X_0 is damaged.

Say the closing is *simple* if a and b are both easy, and *partnered* otherwise; put $\varepsilon = 0$ in the first case and $\varepsilon = 1$ in the second.

Lemma 4.1. $X_0 \setminus \{a, b\}$ contains at least $2h + 2\varepsilon$ easy elements.

Proof. By Corollary 2.5(b) and $|O'| \geq n/6$, the set $O' \setminus \{a, b\}$ contains at least $n/6 - 0.1578n - 2 \geq 0.0088n - 2$ easy elements. If $h = 0$ we need at most 2 of them, and since $l - 1 \geq 2$ and easy elements are listed first, X_0 contains two. If $h > 0$, then X_0 contains a hard element of the list, so it contains every element listed before it, in particular all easy elements of $O' \setminus \{a, b\}$; and $0.0088n - 2 \geq 2 \cdot 5.82 \cdot 10^{-4}n + 2 \geq 2h + 2$. $\square$

Using Lemma 4.1, choose distinct easy elements of $X_0 \setminus \{a, b\}$: two elements y_a, y_b if the closing is partnered, and two elements $y_{i,1}, y_{i,2}$ for each $i = 1, \dots, h$ (“partners”). Let R be the set of elements of X_0 that are neither a, b , a hard z_i , nor a partner. Every element of R is ordinary (non-hard).

4.2. Choice of the internal connectors. We now choose distinct elements of $E^* \cap A$, collected in a set U_c , greedily.

Step 0. If the closing is partnered, choose $w_a \in N^*(\text{lcm}(y_a, a)) \cap A$ and $w_b \in N^*(\text{lcm}(b, y_b)) \cap A$ with $w_a \neq w_b$.

Step i ($1 \leq i \leq h$). Choose $w_{i,1} \in N^*(\text{lcm}(y_{i,1}, z_i)) \cap A$ and $w_{i,2} \in N^*(\text{lcm}(z_i, y_{i,2})) \cap A$, distinct from each other and from all previously chosen connectors.

Lemma 4.2. All these choices are possible. Consequently $|U_c| = 2h + 2\varepsilon \leq 1.164 \cdot 10^{-3}n + 2$.

Proof. All least common multiples involved are at most n^2 , and $a, b, z_i \in O'$, so (6) applies. For step 0 it gives at least $\frac{n}{8}\rho'(a) - \sqrt{n} \geq \frac{n}{8}r(n) - \sqrt{n} \geq 2$ candidates for w_a by (4), and likewise for w_b . For step i write $x = \rho'(z_i) < 0.56$; each of the two candidate sets has at least $\frac{n}{8}x - \sqrt{n}$ elements. The connectors chosen before step i number at most $2 + 2(i - 1)$, and $i - 1$ is at most the number of $z \leq n$ with $\rho'(z) \leq x$, which by Corollary 2.5(a) is at most $1.04 \cdot 10^{-3}xn$. Hence at least

$$\frac{n}{8}x - \sqrt{n} - 2 - 2.08 \cdot 10^{-3}xn - 1 \geq 0.1229xn - \sqrt{n} - 3 \geq 0.1229r(n)n - \sqrt{n} - 3 > 1$$

candidates remain for each of $w_{i,1}, w_{i,2}$ (the extra -1 accounts for $w_{i,1}$ when choosing $w_{i,2}$), using $x \geq r(n)$ and n large. $\square$

4.3. Super-vertices and the auxiliary bipartite graph. Define the following paths in $G(A)$; each is a path because each listed connector lies in N^* of the least common multiple of its two neighbours, and $a \sim b$:

$$S_0 = \begin{cases} a, b & \text{(simple closing),} \\ y_a, w_a, a, b, w_b, y_b & \text{(partnered closing),} \end{cases} \quad S_i = y_{i,1}, w_{i,1}, z_i, w_{i,2}, y_{i,2} \quad (1 \leq i \leq h).$$

These paths are vertex-disjoint, their odd vertices are exactly the elements of $X_0 \setminus R$, and their even vertices are exactly U_c . Each path S has two ends p_S, q_S (its first and last vertex), both easy (in the simple case the ends a, b are easy by definition). Hence

$$(7) \quad \rho'(\text{lcm}(p_S, q_S)) \geq 0.75^2 = 0.5625.$$

Let $X' = R \cup \{S_0, S_1, \dots, S_h\}$, where each S_i is regarded as a single new vertex, and let $B' = (E^* \cap A) \setminus U_c$. Let $\mathcal{H}$ be the bipartite graph with parts X' and B' in which $x \in R$ is joined to $\beta \in B'$ if $\beta \in N^*(x)$, and $S \in \{S_0, \dots, S_h\}$ is joined to β if $\beta \in N^*(\text{lcm}(p_S, q_S))$.

Lemma 4.3. Let $k = \lceil 0.170n \rceil$. Then every vertex of X' has at least k neighbours in $\mathcal{H}$, $2 \leq |X'| \leq k$, and $k \leq |B'| \leq 2k - 3$.

Proof. For $x \in R$ we have $\rho'(x) \geq 0.56$, and for a super-vertex (7) holds. By Lemma 2.2, at most $d_E \leq n/70$ elements of $N^*(\cdot)$ are missing from A , and at most $|U_c|$ are used as connectors, so each vertex of X' has at least

$$\frac{n}{3} \cdot 0.56 - \sqrt{n} - \frac{n}{70} - |U_c| \geq (0.18666 - 0.01429 - 0.00117)n - \sqrt{n} - 2 \geq 0.1712n - \sqrt{n} - 2 \geq k$$

neighbours in B' . Next, $|X'| \leq |X_0| = l + 1 \leq o(n) + 1 \leq n/6 + 1 \leq k$. Counting, $|X'| = (l + 1) - (2 + h + 2h + 2\varepsilon) + (1 + h) = l - 2h - 2\varepsilon$. If $h = 0$ this is at least $l - 2 \geq 3$. If $h > 0$ then, as in the proof of Lemma 4.1, X_0 contains every non-hard element of O' , so $l + 1 \geq |O'| - 5.82 \cdot 10^{-4}n \geq n/6 - 5.82 \cdot 10^{-4}n$ and $|X'| \geq n/6 - 1.8 \cdot 10^{-3}n - 3 \geq 2$. Finally $|B'| \leq |E^*| \leq n/3 + 1 \leq 0.34n - 3 \leq 2k - 3$, and $|B'| \geq n/3 - 1 - n/70 - |U_c| \geq 0.317n \geq k$. $\square$

4.4. Conclusion. By Lemma 4.3 and Theorem 2.1, $\mathcal{H}$ contains a cycle C of length $2|X'|$ which passes through every vertex of X' and alternates between X' and B' . For each super-vertex S on C , let $\beta, \beta' \in B'$ be its two neighbours on C . Both lie in $N^*(\text{lcm}(p_S, q_S))$, hence β is coprime to p_S and β' is coprime to q_S . Replace the segment β, S, β' of C by the path β, S, β' in which S is now traversed from p_S to q_S . Every $x \in R$ on C is adjacent in $G(A)$ to its two neighbours on C , which lie in $N^*(x)$.

The result is a closed walk in $G(A)$: its vertices are elements of $B' \cup U_c \subseteq E^* \cap A$ and of $X_0 \subseteq O \subseteq A$. It has no repeated vertex, because the paths S_i are disjoint from each other, from R , and from B' (their even vertices form U_c , which was removed). Hence it is a cycle of $G(A)$, of length

$$2|X'| + \sum_{i=0}^h (|S_i| - 1) = 2(l - 2h - 2\varepsilon) + (1 + 4\varepsilon) + 4h = 2l + 1,$$

since $|S_0| = 2 + 4\varepsilon$ and $|S_i| = 5$ for $i \geq 1$. Together with the short cycles of Section 3, this proves Theorem 1.1. $\square$

5. THE CONDITIONS ON n

For the reader's convenience we list every place where n was assumed large; each is an explicit inequality between elementary functions of n , so n_0 is effective.

- (1) $\log(2 \log n) \geq 3.84$, giving $2^{\omega(m)} \leq \sqrt{n}$ for $m \leq n^2$ in (3);
- (2) $e^\gamma(\log \log n + \log 2) + 2.51/\log \log(n^2) \leq 2 \log \log n + 3$, giving (4);
- (3) $n \geq 13$ (Lemma 2.6), $n \geq n_1$ and $cn \geq 4$ with c, n_1 from [2, Theorem 1] (Section 3);
- (4) $11n/70 + 3 + 3.09 \cdot 10^{-4}n \leq 0.1578n$ (Corollary 2.5(b));
- (5) $0.0088n - 2 \geq 1.164 \cdot 10^{-3}n + 2$ (Lemma 4.1);
- (6) $0.1229r(n)n \geq \sqrt{n} + 5$ and $\frac{n}{8}r(n) \geq \sqrt{n} + 2$ (Lemma 4.2);
- (7) $0.1712n - \sqrt{n} - 2 \geq \lceil 0.170n \rceil$, $n/6 + 1 \leq 0.170n$, $n/3 + 1 \leq 0.34n - 3$ and $0.3175n - 3 \geq \lceil 0.170n \rceil$ (Lemma 4.3).

6. WHAT REMAINS FOR QUESTION 1

The hypothesis $|E^* \setminus A| \leq n/70$ is used in exactly two places: in (6) (to bound $a(z)$ for non-hard z) and in Lemma 4.3, where every vertex of X' must keep more than half of B' and at least $|X'| \approx n/6$ neighbours. If A omits a set $D \subseteq E^*$ of size δn , these requirements become $\frac{n}{3}\rho'(x) - \delta n > \frac{1}{2}(\frac{n}{3} - \delta n)$ and $\frac{n}{3}\rho'(x) - \delta n \geq \frac{n}{6}$ in the worst case, where all of D lies in the neighbourhood of x . For δ of order $1/10$ and the longest cycles (l close to $n/6$) they fail even for easy x , and once $|E^* \cap A| < n/6$ no cycle alternating between odd numbers and elements of E^* can contain $n/6$ odd numbers at all. In that regime the surplus $|O| - o(n) - 1 \geq |D|$ of odd elements must be used, and the cycles must contain edges between odd numbers (for instance between an odd multiple of 3 and an element of A coprime to 6). We have not found an argument controlling these edges for an arbitrary set of surplus odd numbers, so Question 1 remains open when $|E^* \setminus A| > n/70$.

CODE AVAILABILITY

The scripts used for the numerical constants of Lemma 2.3 (`constants.py`), the exhaustive check of Question 1 for $n \leq 21$ (`bruteforce_small_n.py`) and the empirical distribution of ρ' (`rho_distribution.py`) are distributed with this paper.

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